NAME : PRADEEPA. S

ROLL NO : 22CSB15

DEPT : CSE

SUBJECT : Engineering Mathematics - II (M_2)

Part - A

1. A first order differential equation is an equation contains the derivation of a function but it contain higher derivatives

Example:- $\frac{dy}{dx} + y = x$

2. $$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

3. $$\frac{dy}{dx} + P(y)x = Q(x)$$

4. The Laplace transform of a function $f(t)$

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt.$$

5. The rank of a matrix is the minimum number of linearly dependent rows or columns in the matrix.

PART-B

6. $\frac{dy}{dx} + y = e^x \rightarrow (1)$

Now, $P(x) = 1$

$Q(x) = e^x$

factor $F = e^{\int P(x)dx}$

$= e^{\int 1 dx}$

$F = e^x \rightarrow (2)$

$(1) \times (2)$

$$e^x \frac{dy}{dx} + e^x y = e^{2x}$$

$$\frac{d}{dx}(e^{2x}) = ye^x$$

Integrating both sides.

$$\int \frac{d}{dx} e^{2x} dx = \int ye^x dx.$$

$$\frac{e^{2x}}{2} + c = ye^x$$

$\div$ by e^x

$$\boxed{y = \frac{e^x}{2} + c}$$

7. $(2xy+y^2)\,dx + (x^2+2xy)\,dy = 0$

$M = 2xy + y^2$

$N = x^2 + 2xy.$

$M = \frac{\partial M}{\partial y}$ $\quad N = \frac{\partial N}{\partial x}$

integrating M with respect to x

$\int (2xy+y^2)\,dx.$

$\Rightarrow x^2y + xy^2 + h(y)$

integrating N with respet to y.

$\frac{d}{dy}(x^2y + xy^2 + h(y))$

$\Rightarrow h'(y) = 1$

$h(y) = c$

$\boxed{xy + xy^2 = C}$

8 $f(t) = te^{at}$

$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt.$

9. $F(s) = \frac{s+1}{(s+1)^2+4}.$

formula : $L\{e^{-at}\cos bt\} = \frac{s+a}{(s+a)^2+b^2}$

when $a = 1$, $b = 2$

$$\boxed{\Rightarrow e^{-t}\cos 3t}$$

10.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

formula $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)^2 - 1 = 0.$$

$$\lambda^2 - 4\lambda + 3 = 0.$$

$$\lambda(\lambda - 2) - 1(\lambda - 2) = 0$$

$$(\lambda - 2)(\lambda - 1) = 0$$

$$\boxed{\lambda = 2, \ \lambda = 1}$$